Monthly Energy Report - 2026-08

Report period: 2026-08 Generated: 2026-08-27 15:51 UTC Rate: Rs 7.00/kWh

System Summary

Total energy (ACTUAL): 511.89 kWh
Average power: 2132.86 W
Peak power: 3895.96 W
Peak timestamp: 2026-08-19 06:16 UTC
Active / available PZEM: 9 / 9

PZEM Summary

PZEM	kWh	PeakW	Anom	Fault	Risk	Fcst	Bill
P1	48.41	413.76	0	0	WATCH	FORECAST	OK
P2	48.24	413.77	0	0	WATCH	FORECAST	OK
P3	48.40	398.11	1	0	WATCH	FORECAST	OK
P4	86.83	713.45	0	0	HIGH	FORECAST	OK
P5	48.25	399.65	0	1	WATCH	FORECAST	OK
P6	48.21	408.56	1	0	WATCH	FORECAST	OK
P7	48.20	399.80	0	0	WATCH	FORECAST	OK
P8	87.03	741.92	0	0	HIGH	FORECAST	OK
P9	48.32	408.18	1	0	WATCH	FORECAST	OK

AI Insights

Anomalies: 3 Faults: 1 Emergencies: 0
Maintenance risk: WATCH:7, HIGH:2
System forecast: No data
System bill: OK

[MEDIUM] PZEM 1 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 05:30-08:00 (median 357 W vs typical 210 W).

[MEDIUM] PZEM 1 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 348 W vs typical 207 W).

[MEDIUM] PZEM 2 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 05:30-08:00 (median 368 W vs typical 205 W).

[MEDIUM] PZEM 2 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 353 W vs typical 207 W).

[MEDIUM] PZEM 3 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 07:00-09:30 (median 365 W vs typical 204 W).

[MEDIUM] PZEM 3 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 06:00-08:30 (median 348 W vs typical 210 W).

[MEDIUM] PZEM 4 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 80 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 4.13 kWh)

[MEDIUM] PZEM 4 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 05:00-07:30 (median 639 W vs typical 375 W).

[MEDIUM] PZEM 4 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:30-08:00 (median 628 W vs typical 375 W).

[MEDIUM] PZEM 5 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 04:00-06:30 (median 367 W vs typical 208 W).

[MEDIUM] PZEM 5 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 351 W vs typical 209 W).

[MEDIUM] PZEM 6 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 05:00-07:30 (median 364 W vs typical 215 W).

[MEDIUM] PZEM 6 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 351 W vs typical 209 W).

[MEDIUM] PZEM 7 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 03:30-06:00 (median 365 W vs typical 203 W).

[MEDIUM] PZEM 7 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 04:00-06:30 (median 349 W vs typical 212 W).

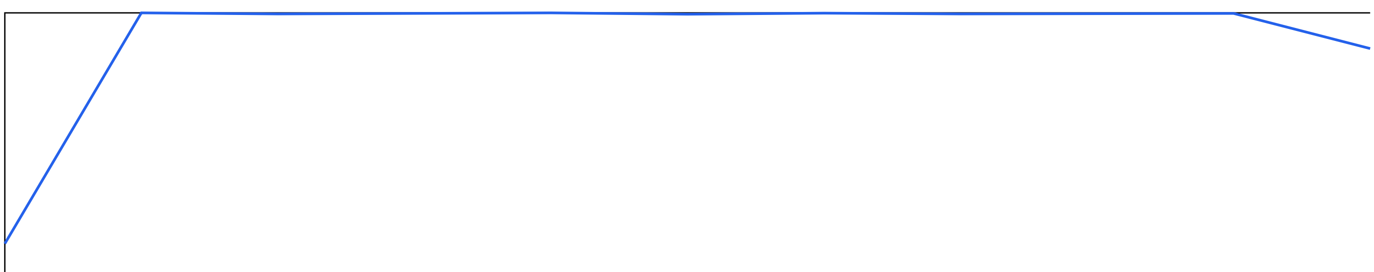
[MEDIUM] PZEM 8 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 80 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 4.05 kWh)
[MEDIUM] PZEM 8 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 04:30-07:00 (median 646 W vs typical 360 W).
[MEDIUM] PZEM 8 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 04:30-07:00 (median 635 W vs typical 373 W).
[MEDIUM] PZEM 9 RESPOND_PREDICTABLE_HIGH_LOAD: Forecast shows a predictable high-load window 05:30-07:30 (median 362 W vs typical 216 W).
[MEDIUM] PZEM 9 SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 355 W vs typical 210 W).
[MEDIUM] SYSTEM SHIFT_NON_CRITICAL_LOAD: Recurring high-demand window 05:00-07:30 (median 3719 W vs typical 2242 W).
[LOW] PZEM 1 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 46 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.40 kWh)
[LOW] PZEM 2 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.21 kWh)
[LOW] PZEM 3 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.19 kWh)
[LOW] PZEM 5 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 43 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.06 kWh)
[LOW] PZEM 6 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.17 kWh)
[LOW] PZEM 7 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 42 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 1.94 kWh)
[LOW] PZEM 9 REDUCE_IDLE_CONSUMPTION: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving. (est 2.17 kWh)

Alert Summary

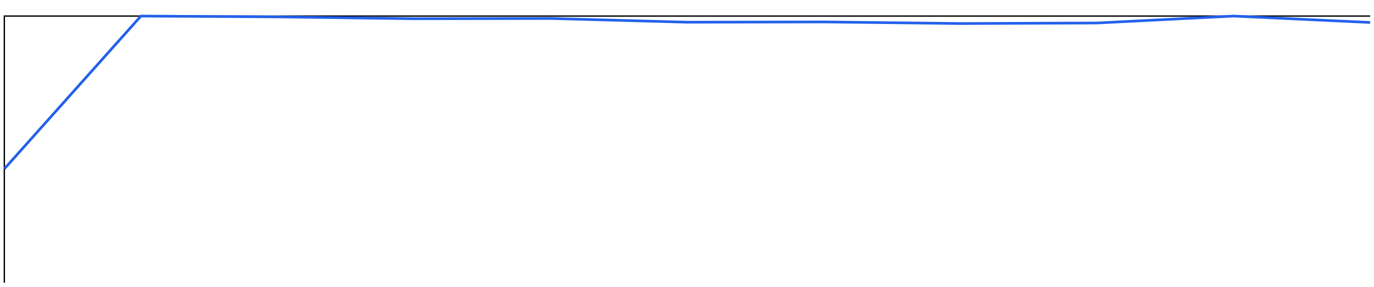
WARNING PZEM 5 (OVER_VOLTAGE) @ 2026-08-26 03:01 UTC

Visual Summary

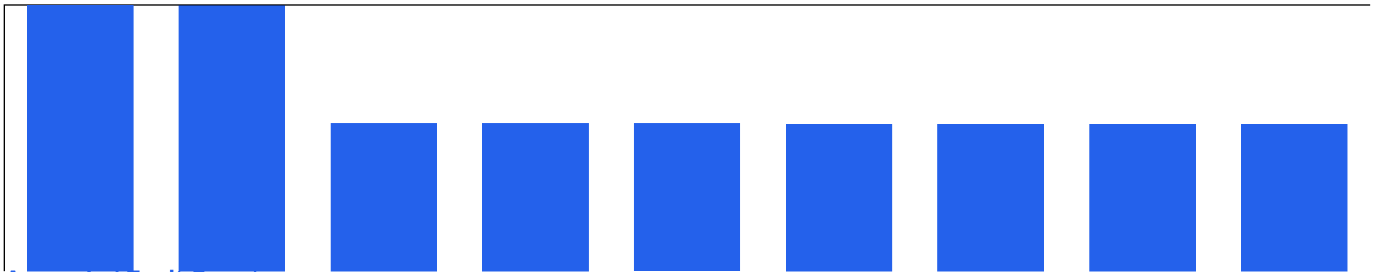
Daily Energy Trend (kWh)



Daily Peak Power (W)



PZEM Energy Comparison (kWh)



Anomaly / Fault Count